
Academic Presentations

A LaTeX Template Using the beamer Class

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Name of the Inviting Institution/Seminar Series

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Outline

- ➊ Introduction
- ➋ Study Design
- ➌ Results
- ➍ Discussion
- ➎ Math “Torture” Test
- ➏ References

Introduction

Introduction 1: Choice of a Reasonable Aspect Ratio

When preparing a presentation, we often do not know whether the native aspect ratio of the projector in the seminar room/lecture hall will be 4:3 or 16:9 (or 16:10).

In this case, it may be a good idea to choose an **intermediate aspect ratio**, see <https://github.com/josephwright/beamer/issues/497>. The idea behind this recommendation is that it minimizes the average loss of available space.

Hence, these templates include a presentation in the **14:9 aspect ratio** (see https://en.wikipedia.org/wiki/14:9_aspect_ratio): while it is imperfect for probably every projector that you will encounter, it is good on average for all of them.

(Please note that $14:9 \doteq 1.556$, which is pretty close to the “officially” recommended $20:13 \doteq 1.5385$.)

*Great Minds Discuss Ideas.
Average Minds Discuss Events.
Small Minds Discuss People.*

—<https://quoteinvestigator.com/2014/11/18/great-minds/>

Background

- Temporal discounting is key concept in economics.
- Normative model: exponential discounting. However, observed decisions are hard to explain (e.g., Dohmen et al. 2012).
- One alternative: the “focusing model” by Köszegi and Szeidl (2013).
- See also Lisi (1995).

Introduction 3

Research Question

- The composition of latex and of typical rubbers is given below.
- Is it true that trees are regularly tapped and the coagulated latex which exudes is collected and worked up into rubber?

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Preview of the Results

- There is no feasible method at present known of preventing the inclusion of the resin of the latex with the rubber during coagulation.
- ⇒ Although the separation of the resin from the solid caoutchouc by means of solvents is possible, it is not practicable or profitable commercially.

Introduction 4: Questions to Structure Your Presentation (1/4)

You may find the following list of questions helpful as a guide to making sure that your presentation covers all important aspects. The questions equally apply, of course, to posters and manuscripts.

Introduction 5: Questions to Structure Your Presentation (2/4)

Your introduction might answer the following questions (not necessarily in this order, but this order will produce a “funnel” structure):

1. What do we already know?
2. What do we not know yet (the “knowledge gap”)?
3. Why is it interesting/relevant to close the gap?
4. **What is the research question? [R]**
5. Why is it interesting to answer the research question? [“No one has done it before”
ist not sufficient ...]
6. How do we contribute to closing the knowledge gap: How do we answer the research question? (Which method do we use?)
7. Why is the chosen method (more) suitable (than alternative methods) for answering the research question?

Introduction 6: Questions to Structure Your Presentation (3/4)

Your conclusion/discussion might answer the following questions (the following order will produce a “reverse funnel” structure):

8. What was the knowledge gap before the current study? [Similar to the introduction.]
9. **How did we contribute to closing that gap, and what did we find? [A]** Which method/dataset did we use, and what are the results? [Recap and take-home message.]
10. How do our results relate to (confirm/contradict/complement/extend) the results from previous and other contemporaneous studies?
11. What part of the gap is still open? (Phrased a bit more negatively: What are the limitations of our approach?)
12. Next steps/avenues for future research: How could we go about closing the remaining knowledge gap (by removing the limitations of our current approach)?

Introduction 7: Questions to Structure Your Presentation (4/4)

John Cochrane might do away with question 12.

See his “Writing Tips for Ph.D. Students,”

https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf.

Introduction 8: More Structuring Advice

In her book *The Little Book of Research Writing*, Varanya Chaubey proposes a simpler list of only three items—the “**RAP method**”:

- The “**R**” stands for “research question.”
- The “**A**” stands for “answer” (to the research question, of course).
- The “**P**” stands for “positioning statement.” (How does our manuscript relate to the literature: Does it corroborate or extend or contradict others’ findings?)

Relating the “RAP” approach to the the list of questions above, question 4 is the “R,” the answer to question 9 is the “A,” and the remaining questions spell out the “P.”

For a quick summary of the book, see https://mauve-porcupine-8992.squarespace.com/s/Chaubey_Research_Writing-bxddd.pdf. See also <https://www.econscribe.org/about> and <https://arxiv.org/pdf/2012.07787>.

Study Design

Study Design 1: Beamer block Environments

Block title example: 0123456789 äöüß ÄÖÜ €. Often finding flowers in official fjords

The block environment. The block environment. The block environment. The block environment. The block environment. Block title example: 0123456789 äöüß ÄÖÜ €. Often finding flowers in official fjords.

An exemplary example

I am the exampleblock environment. Use me for examples.

Summary: Things to remember

The alertblock environment. Use this environment for really important stuff. The alertblock environment.

Study Design 2: Beamer block Environment with Different Colors

A block in the default color

The block environment. The block environment. The block environment. The block environment. A block in the default color.

A block in yellow

The block environment. The block environment. The block environment. The block environment. A block in yellow.

A block in the default color

The block environment. The block environment. The block environment. The block environment. A block in the default color.

Study Design 3: Beamer definition and theorem Environments

Definition (A Very, Very, Very, Very, Very, Very Long Name of a Concept that Spans Two Lines)

The definition environment. Upright.

Theorem (Theorem's Name)

The theorem environment. Italic.

Lemma (Lemma's Name)

The lemma environment. Italic.

Corollary (Corollary's Name)

The corollary environment. Italic.

Proof of Theorem's Name

The proof environment. Upright.



Study Design 4: Design of the Study

- The latex of the best rubber plants furnishes from 20% to 50% of rubber.
- As the removal of the impurities of the latex is one of the essential points to be aimed at, it was thought that the use of a centrifugal machine to separate the caoutchouc as a cream from the watery part of the latex would prove to be a satisfactory process.

Study Design 5: Design of the Study

The watery portion of the latex soaks into the trunk, and the soft spongy rubber which remains is kneaded and pressed into lumps or balls:

$BAL_{1:1}^I, BAL_{1:1}^{II}$: Each payment transferred on single day.

$UNBAL_{1:n}^I$: Earlier payoff concentrated, while later payoff dispersed over $n = 2, 4$, or 8 dates.

$UNBAL_{n:1}^{II}$: Earlier payoff dispersed over $n = 2, 4$, or 8 dates, while later payoff concentrated.

Study Design 6: Control Experiment

- Control for alternative explanations.
- Many of the example sentences were taken from <http://sentence.yourdictionary.com/latex>.

Study Design 7: An Example enumerate List

1. First itemtext
 - a. First itemtext
 - i. First itemtext
 - ii. Second itemtext
 - iii. Last itemtext
 - iv. First itemtext
 - b. Second itemtext
 - c. Last itemtext
 - d. First itemtext
2. Second itemtext
3. Last itemtext
4. First itemtext

Study Design 8: An Example `itemize` List

- First itemtext
 - First itemtext
 - First itemtext
 - Second itemtext
 - Last itemtext
 - First itemtext
 - Second itemtext
 - Last itemtext
 - First itemtext
- Second itemtext
- Last itemtext
- First itemtext

Study Design 9: Some Example Text

Let's include some Greek letters: α , β , σ

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper. (a, α) , β , σ , (v, v) , (y, γ)

Test: $\chi\rho\sigma\tau$ $\chi\rho\varsigma$ τ $\chi\rho\sigma$ τ . Math mode, upright: σ

Study Design 10: Some Example Formulas

Let's include some additional Greek letters: $\gamma, \phi, \sigma_\epsilon, c^\alpha$

$$p(R, \phi) \sim \int_{-\infty}^{\infty} \frac{\tilde{W}_n(\gamma) \exp \left[iR/a \left(\sqrt{k^2 a^2 - \gamma^2} \cos \phi \right) \right]}{(k^2 a^2 - \gamma^2)^{3/4} H_n^{(1)} \left(\sqrt{k^2 a^2 - \gamma^2} \right)} d\gamma$$

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Let's include upright Latin letters in math mode: **d, e** (next slide)

$$\int_a^b f(x) dx = F(b) - F(a)$$

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$$\int_a^b f(x) dx = F(b) - F(a)$$

Let's test the math bold styles (both upright and italic)

$$\boldsymbol{\Sigma} := \mathbf{Cov}(\mathbf{X}) = \begin{bmatrix} \text{Var}(X_1) & \cdots & \text{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \cdots & \text{Var}(X_n) \end{bmatrix}$$

Study Design 11: Additional Example Formulas (with upright π)

Only variables are set in italics according to ISO style—hence, we use upright “d,” “e,” and “ π ” (`\mathup{d}`, `\mathup{e}`, and `\mathup{\pi}`, respectively).

Theorem (simplest form of the *Central Limit Theorem*)

Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables with mean 0 and variance 1 on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then

$$\mathbb{P}\left(\frac{X_1 + \dots + X_n}{\sqrt{n}} \leq y\right) \rightarrow \mathfrak{N}(y) := \int_{-\infty}^y \frac{e^{-v^2/2}}{\sqrt{2\pi}} \, dv \quad \text{as } n \rightarrow \infty,$$

or, equivalently, letting $S_n := \sum_1^n X_k$,

$$\mathbb{E}f(S_n/\sqrt{n}) \rightarrow \int_{-\infty}^{\infty} f(v) \frac{e^{-v^2/2}}{\sqrt{2\pi}} \, dv \quad \text{as } n \rightarrow \infty, \text{ for every } f \in \mathcal{BC}(\mathbb{R}).$$

Study Design 12: An siunitx Example Table

Table 1. Overview of the choice lists presented to subjects (adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017).

	Alternative A				Alternative B			
	$C_{A,1}$	$P_{A,1}$	$C_{A,2}$	$P_{A,2}$	$C_{B,1}$	$P_{B,1}$	$C_{B,2}$	$P_{B,2}$
<i>Choice List I: risky/risky ($x = €22.00$, $r = €7.50$, $h = €11.50$; 25 rows)</i>								
Top row	€ 3.00	50%	€22.00	50%	€ 3.00	50%	€ 7.00	50%
Center row	€ 3.00	50%	€22.00	50%	€ 9.00	50%	€13.00	50%
Row with $m = 0$	€ 3.00	50%	€22.00	50%	€10.50	50%	€14.50	50%
Bottom row	€ 3.00	50%	€22.00	50%	€15.00	50%	€19.00	50%
<i>Choice List II: safe/risky ($x = €16.00$, $r = €5.00$, $h = €5.00$; 19 rows)</i>								
Top row	€11.00	100%			€11.00	50%	€21.00	50%
Center row	€11.00	100%			€ 6.50	50%	€16.50	50%
Row with $m = 0$	€11.00	100%			€ 6.00	50%	€16.00	50%
Bottom row	€11.00	100%			€ 2.00	50%	€12.00	50%
<i>Choice List III: "long shot" ($x = €14.00$, $r = -€36.00$, $h = €7.00$; 21 rows)</i>								
Top row	€ 7.00	90%	€50.00	10%	€ 7.00	90%	€10.00	10%
Row with $m = 0$	€ 7.00	90%	€50.00	10%	€11.00	90%	€14.00	10%
Center row	€ 7.00	90%	€50.00	10%	€12.00	90%	€15.00	10%
Bottom row	€ 7.00	90%	€50.00	10%	€17.00	90%	€20.00	10%
<i>Choice List IV: delayed payoffs ($x = €18.00$, $r = €6.00$, $h = €8.50$, paid in one week; 20 rows)</i>								
Top row	€ 9.50	50%	€12.00	50%	€ 9.50	50%	€24.00	50%
Above-center row	€ 9.50	50%	€12.00	50%	€ 5.00	50%	€19.50	50%
Below-center row	€ 9.50	50%	€12.00	50%	€ 4.50	50%	€19.00	50%
Row with $m = 0$	€ 9.50	50%	€12.00	50%	€ 3.50	50%	€18.00	50%
Bottom row	€ 9.50	50%	€12.00	50%	€ 0.00	50%	€14.50	50%

Results

Results 1: Overview

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 - a. In some cases (Allium, Convolvulaceae, etc.) rows of cells with latex-like contents occur.
 - b. However, the walls separating the individual cells do not break down.
2. The rows of cells from which the laticiferous vessels are formed can be distinguished (6.3 p.p. vs. 2.6 p.p.; $p < 0.01$).

Results 2: Our Main Results

The charts are taken from Dertwinkel-Kalt et al. (2017).

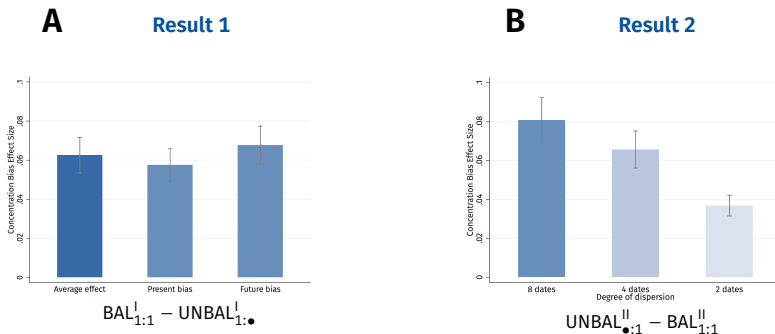
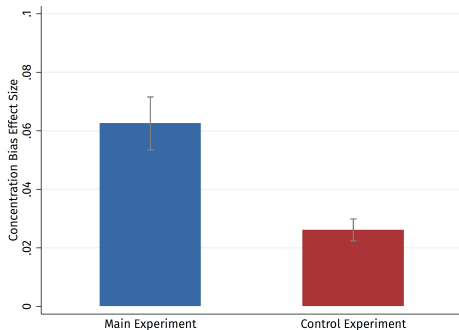


Figure 1. (A) Difference between treatment and control condition. (B) Heterogeneity.

Results 3: Main vs. Control Experiment

Rule out some alternative explanations (Dertwinkel-Kalt et al. 2017).

Result 3



Results 4: Another `siunitx` Example Table

Table 2. Example of a regression table (adapted from Gerhardt, Schildberg-Hörisch, and Willrodt 2017).
Make sure to mention the dependent variable (here, $m_{\sim}$)!

	(1)	(2)	(3)	(4)	(5)
Treatment	-0.390 (+0.352)	-0.228 (-0.205)	-0.729* [+0.377]	-0.449* [-0.245]	-0.453** {+0.204}
Female	0.948*** (0.354)	0.061 (0.233)	0.188 (0.372)	0.305 (0.226)	0.385* (0.222)
Female $\times$ Treatment	0.169 (0.514)	0.251 (0.325)	0.892* (0.533)	0.454 (0.341)	0.439 (0.307)
Final high school grade	-0.101 (0.198)	0.013 (0.144)	0.076 (0.224)	0.117 (0.146)	0.039 (0.133)
Trait self-control	-0.016 (0.016)	0.002 (0.010)	-0.016 (0.015)	0.000 (0.010)	-0.007 (0.009)
Constant	2.357*** (0.239)	1.512*** (0.144)	-0.322 (0.265)	2.158*** (0.161)	1.437*** (0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment $\times$ (1 + Female)	-0.221	0.023	0.163	0.004	-0.014
$p_F[\text{Treatment} \times$ (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Notes: Dependent variable: $m_{\sim}$. Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

Results 5: Revealing a Table Column by Column

Table 3. Revealing a table column by column makes it easier to comprehend.

	(1)	(2)	(3)	(4)	(5)
Treatment					
Female					
Female $\times$ Treatment					
Final high school grade					
Trait self-control					
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Observations					
R^2					
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Observations	303	289	295	304	
R^2	0.057	0.008	0.039	0.043	
Treatment × (1 + Female)	-0.221	0.023	0.163	0.004	
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Results 6: Highlighting Cells in a Table

Table 4. *tabularray* makes it simple to highlight cells, rows, or columns in a table.

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	(+0.352)	(-0.205)	[+0.377]	[-0.245]	{+0.204}
Female	0.948***	0.061	0.188	0.305	0.385*
	(0.354)	(0.233)	(0.372)	(0.226)	(0.222)
Female × Treatment	0.169	0.251	0.892*	0.454	0.439
	(0.514)	(0.325)	(0.533)	(0.341)	(0.307)
Final high school grade	-0.101	0.013	0.076	0.117	0.039
	(0.198)	(0.144)	(0.224)	(0.146)	(0.133)
Trait self-control	-0.016	0.002	-0.016	0.000	-0.007
	(0.016)	(0.010)	(0.015)	(0.010)	(0.009)
Constant	2.357***	1.512***	-0.322	2.158***	1.437***
	(0.239)	(0.144)	(0.265)	(0.161)	(0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment × (1 + Female)	-0.221	0.023	0.163	0.004	-0.014
$p_F[\text{Treatment} \times$ (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Results 6: Highlighting Cells in a Table

Table 4. *tabularray* makes it simple to highlight cells, rows, or columns in a table.

	(1)	(2)	(3)	(4)	(5)
Treatment	-0.390 (+0.352)	-0.228 (-0.205)	-0.729* [+0.377]	-0.449* [-0.245]	-0.453** {+0.204}
Female	0.948*** (0.354)	0.061 (0.233)	0.188 (0.372)	0.305 (0.226)	0.385* (0.222)
Female × Treatment	0.169 (0.514)	0.251 (0.325)	0.892* (0.533)	0.454 (0.341)	0.439 (0.307)
Final high school grade	-0.101 (0.198)	0.013 (0.144)	0.076 (0.224)	0.117 (0.146)	0.039 (0.133)
Trait self-control	-0.016 (0.016)	0.002 (0.010)	-0.016 (0.015)	0.000 (0.010)	-0.007 (0.009)
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Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment $\times$ (1 + Female)	-0.221	0.023	0.163	0.004	-0.014
p_F [Treatment $\times$ (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Results 7: Yet Another `siunitx` Example Table

Table 5. Figure grouping via `siunitx` in a table.

(1)	(2)	(3)
-0.100*	-0.100 01*	-123 456.444***
(2.871)	(2.871 23)	[+50 000.123]

Discussion

Discussion 1

- The latex exhibits a neutral, acid, or alkaline reaction, depending on the plant from which it was obtained.

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- The latex exhibits a neutral, acid, or alkaline reaction, depending on the plant from which it was obtained.
- The latex is therefore usually allowed to coagulate on the tree (Kőszegi and Szeidl 2013).
 - ⇒ The latex, which is usually coagulated by standing or by heating, is obtained from incisions.

Discussion 1

- The latex exhibits a neutral, acid, or alkaline reaction, depending on the plant from which it was obtained.
- The latex is therefore usually allowed to coagulate on the tree (Kőszegi and Szeidl 2013).
 - ⇒ The latex, which is usually coagulated by standing or by heating, is obtained from incisions.
- See also Dohmen et al. (2012), Bordalo, Gennaioli, and Shleifer (2013), and Bursztyn et al. (2025).

Discussion 2: Conclusion

A paragraph before a list.

- When exposed to air, the latex gradually undergoes putrefactive changes accompanied by coagulation.

A paragraph within a list item.

- The addition of a small quantity of ammonia or of formalin to some latices has the effect of preserving them.
- There is, however, reason to believe the following.
- The coagulation of latex into rubber is not mainly of this character.

A paragraph after a list.

Discussion 3: An Automated Animation

The automated transition to the next slide (= page in the PDF document) only works in full-screen mode.

- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 1—Angle: 30.0°

Discussion 3: An Automated Animation

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Figure 2. Step 2—Angle: 60.0°

Discussion 3: An Automated Animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 3—Angle: 90.0°

Discussion 3: An Automated Animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 4—Angle: 120.0°

Discussion 3: An Automated Animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 5—Angle: 150.0°

Discussion 3: An Automated Animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 6—Angle: 180.0°

Discussion 3: An Automated Animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 7—Angle: 210.0°

Discussion 3: An Automated Animation

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- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 8—Angle: 240.0°

Discussion 3: An Automated Animation

The automated transition to the next slide (= page in the PDF document) only works in full-screen mode.

- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 9—Angle: 270.0°

Discussion 3: An Automated Animation

The automated transition to the next slide (= page in the PDF document) only works in full-screen mode.

- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 10—Angle: 300.0°

Discussion 3: An Automated Animation

The automated transition to the next slide (= page in the PDF document) only works in full-screen mode.

- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.



Figure 2. Step 11—Angle: 330.0°

Discussion 3: An Automated Animation

The automated transition to the next slide (= page in the PDF document) only works in full-screen mode.

- The feature is available in Adobe Acrobat and Acrobat Reader.
- Unfortunately, it is (currently, December 16, 2025) not available in macOS Preview, Skim, and SumatraPDF.

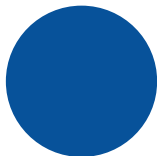


Figure 2. Step 12—Angle: 360.0°

Discussion 4: Testing the `allowframebreaks` option

Let's test automatic numbering with the `allowframebreaks` option.

On this slide, **no** number should be included in the frame title.

Discussion 5: Testing the allowframebreaks Option (1/3)

Let's test automatic numbering with the allowframebreaks option.

On this slide, **“(1/3)”** should appear in the frame title.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Discussion 6: Testing the `allowframebreaks` Option (2/3)

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Discussion 7: Testing the `allowframebreaks` Option (3/3)

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Math “Torture” Test

Math “Torture” Test (1/13)

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathfrak{f}, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F, —$
 $g, \gamma, \Gamma, g, \gamma, \Gamma, g, \gamma, \Gamma, g, \gamma, `$

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathfrak{f}, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F, —$
 $g, \gamma, \Gamma, g, \gamma, \Gamma, g, \gamma, \Gamma, g, \gamma, `$

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathfrak{f}, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F, —$
 $g, \gamma, \Gamma, g, \gamma, \Gamma, g, \gamma, \Gamma, g, \gamma, `$

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, \mathfrak{f}, A, B, \Gamma, \Delta, E, Z, H, \Theta, I, K, \Lambda, M, N, \Xi, O, \Pi, P, \Sigma, T, \Upsilon, \Phi, X, \Psi, \Omega, F, —$
 $g, \gamma, \Gamma, g, \gamma, \Gamma, g, \gamma, \Gamma, g, \gamma, `$

Math “Torture” Test (2/13)

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \Gamma, \Delta, \text{E}, \text{Z}, \text{H},$
 $\Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F}, - \text{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \Gamma, \Delta, \text{E}, \text{Z}, \text{H},$
 $\Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F}, - \text{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \Gamma, \Delta, \text{E}, \text{Z}, \text{H},$
 $\Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F}, - \text{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma$

$\alpha, \beta, \gamma, \delta, \varepsilon, \varepsilon, \zeta, \eta, \theta, \theta, \iota, \kappa, \lambda, \mu, \nu, \xi, \omicron, \pi, \pi, \rho, \rho, \sigma, \varsigma, \tau, \upsilon, \phi, \phi, \chi, \psi, \omega, \text{f}, \text{A}, \text{B}, \Gamma, \Delta, \text{E}, \text{Z}, \text{H},$
 $\Theta, \text{I}, \text{K}, \Lambda, \text{M}, \text{N}, \Xi, \text{O}, \Pi, \text{P}, \Sigma, \text{T}, \text{Y}, \Phi, \text{X}, \Psi, \Omega, \text{F}, - \text{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma, \textbf{g}, \gamma, \Gamma$

Math “Torture” Test (3/13)

Most of the following examples are taken from *The T_EXbook* (Knuth 1984, see <https://ctan.org/pkg/texbook>) and were adapted for L^AT_EX from Karl Berry’s torture test for plain T_EX math fonts.

$$x + y - z, \quad x + y * z, \quad z * y / z, \quad (x + y)(x - y) = x^2 - y^2,$$

$$x \times y \cdot z = [xy]z, \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,$$

$$x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y / z, \quad x := y, \quad x \leq y \neq z, \quad x \sim y \simeq z \quad x \equiv y \not\equiv z, \quad x \subset y \subseteq z$$

$$\sin 2\theta = 2 \sin \theta \cos \theta, \quad O(n \log n \log n), \quad \Pr(X > x) = \exp(-x/\mu),$$

$$(x \in A(n) \mid x \in B(n)), \quad \bigcup_n X_n \parallel \bigcap_n Y_n$$

$$\text{In-text matrices } \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} a & b & c \\ 1 & m & n \end{pmatrix}.$$

Math “Torture” Test (4/13)

Disambiguation: 0 O O, 1 l l | l l /, i j, r n m, θ Θ , ϕ ψ , – –

Latin vs. Greek: a α , d δ , e ϵ , i ι , k κ , n η , o σ , p ρ , β β , u v , v v , w ω , x χ , y γ ,
 A Δ Λ , O Θ Ω , T Γ , Y Υ .

Disambiguation: 0 O O, 1 l l | l l /, i j, r n m, θ Θ , ϕ ψ , – –

Latin vs. Greek: a α , d δ , e ϵ , i ι , k κ , n η , o σ , p ρ , β β , u u , v v , w ω , x χ , y γ , A Δ Λ , O Θ Ω ,
 T Γ , Y Υ .

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x} \frac{1}{1-x^2} = \frac{a+1}{b} \Big/ \frac{c+1}{d}.$$

$$\sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + x}}}}}$$

$$\sqrt[n]{1 + \sqrt[k]{1 + \sqrt[5]{1 + \sqrt[4]{1 + \sqrt[3]{1 + x}}}}}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |\phi(x + iy)|^2 = 0$$

$$\pi(n) = \sum_{m=2}^n \left[\left(\sum_{k=1}^{m-1} \lfloor (m/k) / \lceil m/k \rceil \rfloor \right)^{-1} \right].$$

$$\int_0^\infty \frac{t - ib}{t^2 + b^2} e^{iat} dt = e^{ab} E_1(ab), \quad a, b > 0.$$

$$\mathbf{A} := \begin{pmatrix} x - \lambda & 1 & 0 \\ 0 & x - \lambda & 1 \\ 0 & 0 & x - \lambda \end{pmatrix}.$$

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \begin{pmatrix} u & x \\ v & y \\ w & z \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

$$\mathbf{M} = \begin{matrix} & C & I & C' \\ \begin{matrix} C \\ I \\ C' \end{matrix} & \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{matrix}$$

$$\sum_{n=0}^{\infty} a_n z^n \text{ converges if } |z| < \left(\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \right)^{-1}.$$

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \rightarrow f'(x) \quad \text{as } \Delta x \rightarrow 0.$$

$$\|u_i\| = 1, \quad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of } \left\{ \begin{array}{l} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{array} \right\} \text{ is } \left\{ \begin{array}{l} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{array} \right\}.$$

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$$\|u_i\| = 1, \quad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of } \left\{ \begin{array}{l} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{array} \right\} \text{ is } \left\{ \begin{array}{l} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{array} \right\}.$$

$$\begin{aligned}T(n) &\leq T(2^{\lceil \lg n \rceil}) \leq c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\&< 3c \cdot 3^{\lg n} \\&= 3c n^{\lg 3}.\end{aligned}$$

$$\begin{aligned}(x + y)(x - y) &= x^2 - xy + yx - y^2 \\&= x^2 - y^2 \\(x + y)^2 &= x^2 + 2xy + y^2.\end{aligned}$$

$$\begin{aligned}\left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} dr d\theta \\ &= \int_0^{2\pi} \left(e^{-\frac{r^2}{2}} \Big|_{r=0}^{r=\infty} \right) d\theta \\ &= \pi.\end{aligned}$$

$$\prod_{k \geq 0} \frac{1}{(1 - q^k z)} = \sum_{n \geq 0} z^n / \prod_{1 \leq k \leq n} (1 - q^k).$$

$$\sum_{\substack{0 < i \leq m \\ 0 < j \leq n}} p(i, j) \neq \sum_{i=1}^p \sum_{j=1}^q \sum_{k=1}^r a_{ij} b_{jk} c_{ki} \neq \sum_{\substack{1 \leq i \leq p \\ 1 \leq j \leq q \\ 1 \leq k \leq r}} a_{ij} b_{jk} c_{ki}$$

Math “Torture” Test (13/13)

$$\max_{1 \leq n \leq m} \log_2 P_n \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Inline math: $\max_{1 \leq n \leq m} \log_2 P_n$ and $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

$$p_1(n) = \lim_{m \rightarrow \infty} \sum_{v=0}^{\infty} (1 - \cos^{2m}(v!^n \pi / n))$$

Inline math: $p_1(n) = \lim_{m \rightarrow \infty} \sum_{v=0}^{\infty} (1 - \cos^{2m}(v!^n \pi / n))$

References

References

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Appendix

Appendix: Modeling Concentration Bias

Subjects consider a sequences of consequences $\mathbf{c}$ from choice set $\mathbf{C}$.

- **Standard discounted utility:** Suppose that the instantaneous utility function u satisfies $u' > 0$ and $u'' \leq 0$, and that earlier consequences are preferred over later consequences of the same magnitude, i.e., $D(t) \leq 1$:

$$U(\mathbf{c}) := \sum_{t=1}^T D(t) u(c_t), \quad \text{where, e.g.,} \quad D(t) = \delta^t \text{ or } D(t) = \frac{1}{1+k t}.$$

- **Focusing model (Kőszegi and Szeidl 2013):**

$$\tilde{U}(\mathbf{c}, C) := \sum_{t=1}^T g_t D(t) u(c_t), \quad \text{where}$$

$$g_t \equiv g[\max_{\mathbf{c}' \in C} D(t) u(c'_t) - \min_{\mathbf{c}' \in C} D(t) u(c'_t)]$$

- Weighting function $g[\cdot]$ increases in difference of maximum and minimum possible utility at a point in time.
- Subjects overweight intertemporal consequences with a greater range.